

Power Estimation

1. Component Identification and Power Calculation

The power board consists of the following components: four DC motors (JGA25-370), two motor drivers (OSOYOO model X), one micro servo motor (SG90), and one ESP32 microcontroller. The electrical power consumed by each component is calculated using the following relation:

$$P = U \times I$$

where U is the supply voltage (in volts) and I is the nominal current drawn by the component (in amperes). The values are taken from the components' datasheets [1][2][3].

Component	Voltage (V)	Estimated Current (A)	Power (W)
DC Motors JGA25-370 (x4)	6 V	$0.5 \times 4 = 2.0$ A	12.00 W
Motor Driver OSOYOO (x2)	5 V	$0.036 \times 2 = 0.072$ A	0.36 W
Micro Servo SG90	5 V	0.700 A	3.50 W
ESP32 Microcontroller	3.3 V	0.200 A	0.66 W
Estimated Total Power	—	~2.97 A	≈ 16.52 W

Table 1: Power estimation per component

2. Battery Sizing

In the mechatronics laboratory, a battery with a capacity of $Q_B = 2$ Ah was selected. Two cells are connected in series, providing a total voltage of 7.4 V. From the estimated total power, the operating current is calculated as:

$$I = P / U = 16.52 / 7.4 \approx 2.23 \text{ A}$$

A safety margin is applied to account for transient overcurrents occurring when a motor stalls (locked-rotor current). By using a derated current of 3 A, the sizing remains conservative and accounts for worst-case operating conditions.

3. Autonomy Verification for a Match

The charge required to power the robot for a match duration of 100 seconds is determined. The electrical charge consumed is given by:

$$Q_M = I \times T = 3 \text{ A} \times (100 / 3600) \text{ h} \approx 83.3 \text{ mAh}$$

The battery capacity ($Q_B = 2,000$ mAh) is significantly greater than the charge required for one match ($Q_M \approx 83.3$ mAh), by a factor of more than 24. It can therefore be concluded that the selected battery is more than sufficient to power the robot throughout an entire match without interruption.

References

[1]OSOYOO Store, "DC Motor Robotic Car Speed Encoder for Arduino/Raspberry Pi." [Online]. Available: <https://osoyoo.store/fr-eu/products/dc-motor-robotic-car-speed-encoder-9v-for-arduino-raspberry-pi-platform-diy>. Accessed: Jun. 4, 2026.

[2]A. Prabhu, "In-Depth: Interface L298N DC Motor Driver Module with Arduino," Last Minute Engineers. [Online]. Available: <https://lastminuteengineers.com/l298n-dc-stepper-driver-arduino-tutorial/>. Accessed: Jun. 4, 2026.

[3]Glarity, "What power supply is needed for the SG90 9g micro servo?" [Online]. Available: <https://askai.glarity.app/search/What-power-supply-is-needed-for-the-SG90-9g-micro-servo>. Accessed: Jun. 4, 2026.